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Survey of Techniques for Pulmonary Disease Classification using Deep Learning

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***Abstract*— The field of medical science is getting more effective with emerging trends of computer science technologies like AI, ML, and DL. The area where these technologies play an important role is the detection or recognition of diseases. This paper discusses existing methodologies and various steps involved in the process of detection/recognition of pulmonary diseases using lung sound. The paper is divided into 4 sections. These sections include general steps of any recognition system using lung sound and study of existing methods. The paper helps to overview the different approaches/experiments and build new and effective ones which can give better accuracy.**

Keywords— Pulmonary Diseases, CNN, MLP, RNN, MFCC,

I. INTRODUCTION

Many deaths are caused by pulmonary disorders, which include Chronic Obstructive Pulmonary Disease (COPD), asthma, pneumonia, bronchitis, and others. If not treated effectively, these diseases can lead to death. When compared to healthy people, individuals with these disorders have different lung sounds. Crackles, rhonchus, and wheezes are all common noises heard by persons who have lung disorders. Differentiating between normal and pathological breath sounds can be done using energy, frequency, pitch, and a variety of other parameters. With a stethoscope, medical practitioners can simply listen to those sounds. However, these are insufficient for a proper diagnosis of the patient. Physicians can better diagnose it by determining the intensity and variation of the noises. Using multiple ML and DL algorithms to detect such variation in sound, we can properly anticipate the sickness and help cure it sooner.

Any method that uses lung sound to recognise, forecast, or detect pulmonary illnesses has some flow in common. It all starts with data collection, which can take the form of recordings or audio clips. The dataset is then preprocessed to eliminate unwanted noises and acquire valuable insights. This also leads to a boost in precision. Then after, preprocessed data is used to extract features. The set of useful features is fed into the chosen model for training, allowing the system to learn and produce an accurate output from the given input. Finally, performance is evaluated depending on the outcomes to determine the accuracy of the proposed model.

The dataset used in these applications has two main commonalities i.e Crackles and wheezes. Wheezes are continuous sounds and can sound like a whistle as air passes in and out of the airways. While crackles are occasional and transient, they occur when air escapes from the fluid. These crackles and wheezes must be recognized from the audio clip recorded with a digital stethoscope or other electrical equipment. The dataset is then divided into a train set (to train the model to provide enough data to predict the outcome) and a test set (to test the model).

Some metrics are considered for the performance evaluation, which produce the overall result of the model and describe how feasible and effective it is. Specificity, accuracy, F1 score, precision, recall, and other criteria are prevalent.

II. LITERATURE REVIEW

A. Data Preprocessing

The quality of the recorded lung sounds is insufficient to extract the crucial elements. As a result, preprocessing is required to eliminate noise and normalize the data. The audio may contain ambient noise, human speech, heartbeats, and sometimes stethoscope tube noise that sounds like crackles, all of which may result in incorrect classifications and, as a result, lower accuracy while increasing False Negatives, which is detrimental in medical applications. Various researchers' methods will be discussed in further depth.

Empirical mode decomposition (EMD) perfectly analyzes the non-linear and non-stationary data of sound waves. EMD is a method that is used for denoising and improving the quality of sound signals. The signals are divided in Intrinsic mode function(IMD) and residuals by EMD. Authors in [3] compared all types of EMF and they came to know that IMF 1 can be used for the identification of diseases as it contains the region of interest.

G. Shanthakumari et.al in [2] studied five different denoising methods which includes LMS, RLS, SRLS and TVD and DWT. The results showed that Discrete wavelet transform (DWT) performs better than other methods. Performance metrics such as SNR and PSNR were higher for DWT. Also, a performance metric such as Cross-Correlation(CC) was found to be 0.97.

D Singh et. al [5] studied different denoising techniques which are FIR, Butterworth, Wavelet, Moving average filter, Savitzky Golay, and Median filter. They implemented all

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these filters on their dataset and evaluated the results and performance based on the signal-to-noise ratio. To have a good result, a signal should have a higher SNR value. The results showed that the wavelet decomposition has higher values of signal-to-noise ratio compared to the other five filters studied and implemented. 83.43dB was the SNR value of wavelet decomposition. The results are in line with the research in [2].

Meng, F et al. in [4] employed three filters. The initial stage was to apply a FIR band-pass filter to attenuate high- and low-frequency noise in sounds. The residual high- and low-frequency noise, including human speech and heart sounds, was reduced using a modified wavelet filter to partition the data into separate frequency regions. In the third phase, an adaptive filter was employed to attenuate the heart sounds. The authors' filters produced good denoising results in operation.

In [1] S.B Shuvu, et. al performed data preprocessing in 2 steps a) Bandpass filtering b) Segmentation. They used a bandpass of 50Hz to 2500Hz with a 6th order Butterworth bandpass filter. They resampled all audio samples to 8kHz. They segmented each audio sample into the duration of 6s with zero paddings. The first 6s duration was used if any respiratory cycle was larger than 6s.

B. Feature Extraction

Feature Extraction is the next step in the process. It is vital to feed ML algorithms with features that lead to the desired output in order to boost their performance. Time-domain features, frequency domain features, and statistical domain features are the three types of features. In the case of audio signal classification, however, it is clear that the frequency domain characteristic MFCC (Mel Frequency Cepstral Coefficient) performs better. MFCC seeks to represent sound frequencies as humans perceive them, with more sensitivity for lower frequencies and lesser sensitivity for higher frequencies.

DFT on time series data is used to obtain the MFCC. After that, Mel frequency warping is used to compute the log. The inverse DFT is then computed, yielding a signal that is not primarily in the time domain.

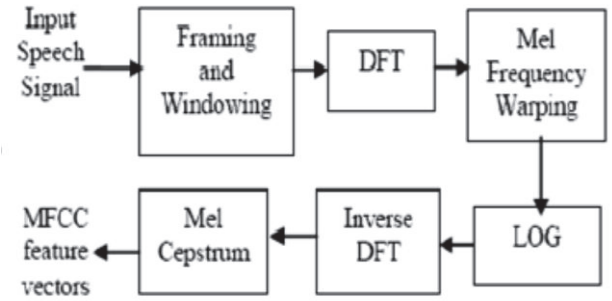


Fig. 1. MFCC extraction

L. Wu., Et al.[6], performed Empirical Mode Decomposition on the samples and produced 3 IMFs (intrinsic mode functions). These IMFs are of the same length as the original sample and can reconstruct the original signal with minimum loss. These three IMFs were used for feature extraction. The Features used were RMS, maximum upper envelope amplitude, mean instantaneous frequency, spectral centroid, spectral flatness, spectral rolloff, and the statistical variation of the spectral features such as mean, median,maximum and minimum were also computed. Along with these, the mean of the 13 MFCCs were also derived. Total 36 features were extracted. They broke the sample into various phases as EIP(extended inspiratory phase), LIP(late inspiratory phase), IP(Balanced Inspiratory Phase) and RC(respiratory cycle). The experiment was designed to understand if a particular respiratory phase has better performance than simply taking the entire RC. The results showed, EIP and IP showed better performance as compared to RC. LIP performed the worst. The EIP and IP based model showed 88% accuracy and 91% precision.

N. H. M. Johari, et al. [10] extracted 13 MFCCs from the samples and computed mean and standard deviation for each frame. These statistical values determine which pattern will provide a distinct outcome between crackles and normal lung sounds by evaluating MFCC coefficients for each segment. Mean reduces the influence of anomalous high values on the output. The results point out that the Standard Deviation of the first three MFCC are useful for identification of the existence of crackles which is supported by the T-test. On the contrary the statistical mean of MFCC is unable to distinguish the crackles and normal sounds.

In [7] D. Perna, Et al. used windows of fixed length for feature extraction. This is very different from other research where respiratory cycles were used. D. Perna used CNN for classification. CNN is majorly used for image classification where the image is a 2D grid of values. He used a 2D matrix of Mel Frequency Cepstral Coefficients which was fed to the CNN. Author developed a Binary and a Ternary classifiers with regulation. Binary classifier which achieved an accuracy of 83% and precision of 96%, while the Ternary Classifier with regulation achieved an accuracy of 87% and precision of 82%.

A year later D. Perna, Et al.[8] came up with another solution where he used RNN for classification. Feature extraction was followed by normalization to avoid getting stuck in the local minima. They used Min-Max normalization and Z-normalization and it was found that Z-normalization provided better prediction. It was found that their RNN-LSTM version S4 and S7 outperformed CNN based approaches. Authors describe it as a result of higher number of features and finer grain windowing used for generating the features.

ResNet-50 was used by H. Chen et al. [9] to classify crackles, wheezes, and normal noises. The method is based on the OST transform, which in the preprocessing stage accentuates the peculiarities of respiratory sounds, and the ResNet overcomes the vanishing gradient problem. For the raw samples, the OST coefficients are computed, and spectrograms are generated, which are then rescaled into three fixed size feature maps of RGB values. These spectrograms are then fed to the ResNet to perform ternary classification. This model provided sensitivity of 96.27%, specificity of 100% and accuracy of 98.79%. Which outperforms traditional ML methods, CNN and ensemble CNN based on MFCCs.

In [11] V. Bansal, et al. used MFCC for classification of the cough sounds. The dataset comprises 501 audio files for training and validation. 2 approaches are used in this paper. The first is to use MFCC as a CNN input, and the second is to use the mel spectrogram as a CNN input. Later the

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VGG16 was also tried on the spectrograms.The classifier scored 70.58% accuracy, 80.95% sensitivity, 60.71% precision and 69.59% F1 score. The accuracy for cough and non cough sounds was above 90% but the same architecture when applied to covid and non covid samples based on spectrograms didn't give satisfactory results.

C. Classification

Jung, et al.[15] gives effective features to increase the performance of DS-CNN. DS-CNN is a depthwise separable convolution neural network model. The research showed that if features of STFT and MFCC are combined and then passed to DS-CNN will give more accuracy than using a single feature and will work more efficiently. The dataset has 12,691 recordings in total. These recordings consist of wheeze, crackle, normal and unknown sounds. All the recordings are preprocessed to be the length of 1.25s. They found that Shrunk DS-CNN gives more accuracy than CNN which is 85.74%.

G. Chambres, et al.[16] the paper divides the whole process into two parts. The first is micro-level, which gives the classification of sounds. Here features are divided into low level, rhythm, SFX, tonal features. But more focus is kept on low level, because low level features are more efficient to give information about wheezes and crackles. To increase accuracy statistical functions are used on these features. Boosted decision trees are used as a classification model. According to the results, efficient detection of wheezes is done by multiclass models. The second is a macro-level, which gives the final patient classification. Due to the variation in the quality of records the performance is nearly about 50% only.

Miguel Angel Fernandez-Granero, et al.[17] Mainly focuses on the early prediction of COPD. For feature extraction, discrete wavelet transform is used. By using these features some statistics values are generated like mean, skewness, kurtosis, standard deviation, and average power of wavelet coefficients. To select the features which are more correlated to the class, a fast correlation-based filter is used; these efficiently contribute in the process of removing unnecessary features known as feature subset selection. For the early predictions of COPD decision tree forest classifier was selected. Accuracy given by the proposed model is 87.8% along with the specificity of 78.1%.

Srivastava A, et al.[18] in the data preprocessing step all audio clips are prepared to be of 20s length. Five features have been calculated like MFCC, chroma_cens, chroma_stft, Mel spectrogram, and constant Q chromagram. Loudness, mask, shift, and speed augmentation techniques were used for upsampling the dataset. The model structure was developed using CNN. From the results, it is observed that MFCC has contributed more than other features to achieve accuracy. MFCC used with CNN gives a sensitivity of 0.92 and 0.92 specificity.

Fraiwan et.al [14], proposed a model using CNN along with BiLSTM (Bidirectional LSTM). The idea behind combining these two models is to reduce the efforts needed for traditional machine learning techniques. In such approaches, external feature extraction is required, whereas in the proposed model CNN gives spatial features and BiLSTM gives temporal features. The dataset of 213 patients is considered for checking the performance of the proposed new technique. This dataset has a total of 1483 recordings. These recordings are of 5s segments. For the preprocessing of data, three steps have been performed. First is 1D wavelet smoothing. In this step, DWT (Discrete wavelet transform) is used instead of CWT (Continuous wavelet transform) because its function doesn't operate on signal continuously resulting in low computational complexity as compared to CWT. The second is Displacement artifact removal. The third is Z Score normalization. For training, a K-fold cross-validation scheme is used with tenfolds. The overall accuracy achieved via this proposed combined model of CNN and BiLSTM is 99.62% with a kappa value of 98.26%.

III. Evaluation Metrics

The methods mentioned are evaluated based on the certain metrics which are standard evaluation metrics for Machine Learning and Deep Learning classification models. These metrics include Accuracy, Precision, Recall, Specificity and Sensitivity.

A. Accuracy (ACC)

It is the ratio of the number of correct predictions to the total number of input samples.

$$\text{Accuracy} = (T_P + T_N) / (T_P + T_N + F_P + F_N)$$

B. Precision (PRE)

Precision is the fraction of True Positives divided by the sum of the True Positives and False positives.

$$\text{Precision} = T_P / (T_P + F_P)$$

C. Recall (REC)

Recall is the fraction of True Positives divided by the sum of the True Positives and False Negatives. Recall is also called Sensitivity.

$$\text{Sensitivity (Recall)} = T_P / (T_P + F_N)$$

D. F1 score

The F-score or F-measure is a measure of a test's accuracy. It is calculated from the precision and recall of the test.

$$\text{F1 Score} = (2 * \text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

E. Specificity (SPE)

Recall is the fraction of True Negatives divided by the sum of the True Negatives and False Positives.

$$\text{Specificity} = T_N / (T_N + F_P)$$

IV. Comparative Study

The comparison for the methodologies will be done on the basis of the type of Neural Network or ML model used. Other comparison criteria include the number of classes and previously mentioned evaluation metrics.

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TABLE I. COMPARISON TABLE

Model	Ref	Method	CC	ACC	PRE	REC	SPE	F1
MLP	19	LPCC based features with MLP	2	0.99	0.99	0.99	0.99	0.99
CNN	7	Binary Classification with MFCC	2	0.83	0.96	0.83	NA	0.88
	7	Ternary Classification with MFCC	3	0.82	0.87	0.82	NA	0.84
	11	Binary classification using MFCC for cough classification as covid positive or negative	2	0.70	0.60	0.80	NA	0.70
	13	Spectrogram operation, parallel pooling structure along with proposed CNN architecture	4	0.72	0.69	0.61	0.86	0.65
	15	STFT, MFCC with DS CNN	4	0.85	0.90	0.90	NA	0.90
	18	MFCC with CNN	2	NA	NA	0.93	0.93	0.93
RNN	8	Separate Classification based on anomaly and Pathological finding using RNN LSTM	6	0.99	0.95	0.92	0.79	0.94
Res Net	9	Ternary classification using Optimized S transform for MFCC extraction and ResNet for solving vanishing gradient problem	3	0.98	NA	0.96	1.00	NA
Hybrid	12	RNN BiLSTM- CNN hybrid model with patient specific model tuning	4	0.96	0.59	0.59	0.84	0.67
	14	1-D Wavelet Smoothing, CNN-RNN BiLSTM	6	0.99	0.98	0.98	0.99	NA
	17	decision tree forest	2	0.88	NA	0.78	0.96	0.80

V. CONCLUSION

The DWT has a higher SNR ratio, which could be due to the wavelet basis's ability to perfectly reconstruct curves with linear and higher order polynomial shapes despite their irregular shape. The Fourier-based filters, on the other hand, do not have this advantage.

So far, the greatest feature for sound signal classification has indeed been MFCC. Normalization techniques and Optimized S-Transform can increase the model's accuracy much more. Another method is to decompose the original sample into IMFs and then extract MFCC for only the necessary IMFs, hence improving the feature quality.

According to this survey, CNN models were primarily employed in the original research, but the limitation is due to the vanishing gradient problem, which can be addressed with ResNets. However, these methods only learn spatial patterns. Audio signals have their own temporal patterns. RNNs and its variants are well-suited for such purposes. It's also been discovered that the model can learn both temporal and spatial features, with excellent results when CNN and RNN are combined

The quality of the dataset is a major hurdle for research in this field. Because of the noise and variance in the dataset, the performance is observed to be impaired. Increasing the size of the dataset can lead to better results coupled with denoising techniques and filter pipeline.

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